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A customized ensemble machine learning approach: predicting students' exam performance

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ABSTRACT

Accurately predicting students' exam performance is crucial for fostering academic success and timely interventions. This study addresses the significant challenge of predicting whether a student will pass or fail based on key factors such as study hours and previous exam scores. Using a dataset of 500 students sourced from Kaggle, we introduce a novel customized ensemble machine learning model, combining Random Forest (RF) and AdaBoost classifiers with a custom-weighted soft voting method (weights of 0.2 for RF and 0.8 for AdaBoost). The model's hyperparameters were optimized via GridSearchCV with 10-fold cross-validation, ensuring robustness. The performance of the ensemble model was evaluated using metrics like Cohen's Kappa, achieving superior predictive accuracy compared to baseline models. Our findings indicate that the proposed model not only improves prediction accuracy but also reduces prediction time, offering practical implications for educators and policymakers to design tailored interventions for at-risk students, ultimately enhancing educational outcomes.

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SUBJECTS

Artificial Intelligence;
Computer Engineering;
Biomedical Engineering

Introduction

Research from Cambridge reveals that students often feel worthless, inadequate, or unintelligent when they fail to obtain high exam scores (Helping Adolescents to Feel Competent and Purposeful – Not Just Happy – May Improve Grades, 2023). This investigation seeks to forecast student exam outcomes (pass or fail) using their prior test results and current study time, employing a tailored ensemble machine learning approach. Higher education institutions are structured to deliver high-quality instruction that can potentially enhance or reduce an individual's awareness, knowledge, or cognitive capabilities. To evaluate teaching effectiveness, educators continuously work to boost student

performance. Modern technological progress allows educators to employ data mining and analytics methods to examine large databases for patterns in student learning and behavior (Kour et al., 2021). While academic achievement is a complex phenomenon affected by various elements, such as the learning environment and personal study practices, it remains a crucial aspect of education. Multiple studies (Baashar et al., 2022; Khan et al., 2022; Song et al., 2021) have created models to predict students' academic performance across various educational stages, including secondary and tertiary levels in different fields, particularly engineering, and medicine, by incorporating numerous indicators or variables.

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Moreover, accurately predicting student performance in exams has long been a significant concern in educational research. However, despite the rapid advancements in machine learning techniques, many existing models fall short in terms of prediction accuracy or generalizability. Institutions continue to struggle with identifying students who are at risk of poor performance in a timely manner. While several models, such as regression-based approaches and traditional machine learning algorithms, have been employed to predict academic outcomes, these methods often fail to capture the complex and non-linear relationships between influencing factors like study hours and previous exam scores.

Recent studies have attempted to address this issue by incorporating more advanced machine learning techniques, but gaps remain. For instance, Alsariera et al. (2022) highlighted that classical machine learning models, while useful, tend to underperform due to low accuracy rates in certain datasets. Additionally, while researchers like Hasan et al. (2020) and Syed Mustapha (2023) focused on feature selection to improve model precision, they did not emphasize optimizing algorithm accuracy. These limitations underscore the need for enhanced models that both reduce computational complexity and increase predictive accuracy. Additionally, the model proposed in this study fills these gaps by introducing a customized ensemble machine learning approach, combining Random Forest (RF) and AdaBoost classifiers with a unique weighted soft voting technique. This approach not only optimizes the predictive power but also significantly reduces prediction time, making it a practical tool for educators to implement. By focusing on key features such as study hours and previous exam scores, the model aims to provide early, accurate predictions that can lead to targeted interventions for students at risk of failure. Thus, the proposed model advances beyond previous works by offering a robust, efficient, and scalable solution for educational institutions.

Furthermore, the factors contributing to a student's performance are complex and interrelated, which limits the efficacy of alternative prediction models such as the regression model, which makes simplistic assumptions about the available data. Nevertheless, certain limitations persist. For instance, Alsariera et al. (2022) conducted a review of studies on various machine-learning algorithms and neural networks. Their research revealed that classic machine-learning algorithms exhibit lower accuracy. Hasan et al. (2020) and Syed Mustapha (2023) focused on optimal feature selection techniques but did not address the optimization of algorithms for accuracy enhancement. Al-Zawqari et al. (2022) employed

random forest and neural networks to predict students' academic performance in online courses. In their study, random forest achieved an 85.83% accuracy score for pass-fail prediction, which presents an opportunity for improvement. In light of these considerations, our proposed method focuses on enhancing the accuracy of student exam performance prediction based on minimal features such as study hours and previous scores. The contributions of this research include the following:

- Students' exam performance prediction utilizing a customized ensemble machine learning model.
- A custom-weighted soft voting method applied to enhance the performance.
- A distinct way to enrich the prediction accuracy of the student's exam performance is through a customized ensemble model for binary classification.
- The proposed model enhanced efficiency by significantly reducing prediction time compared to existing baseline models while maintaining competitive performance.

The study aims to propose a custom ensemble machine learning model that will predict student outcomes based on previous exam scores and current study hours and has the potential to benefit a diverse range of stakeholders. Educational institutions can utilize the predictive model to identify at-risk students and design tailored interventions to improve academic success. Students themselves may benefit from personalized support and knowledge of effective study habits, while parents can use the information to actively support their child's education.

The subsequent sections of this investigation are structured as follows. The resources and methodologies employed to achieve the study's objectives are presented in Section Material and Method. Results and a discussion of the findings are presented in Section Result and Discussion, which concludes with a section on potential directions for future research.

Related studies

Many researchers devote their efforts to predicting the performance of students. In a recent study, Alsariera et al. (2022) investigated the utility of machine learning techniques in this regard. The authors conducted a comprehensive analysis of various machine learning methods and their precision in predicting students' performance. The investigation encompassed Decision Trees (DT), Artificial Neural Networks (ANN), Support Vector Machines, K-nearest neighbors, Linear Regression, and Naive Bayes. The researchers discovered

an accuracy rate of 98.3%, while DT followed closely with an accuracy rate of 98.2%. However, the study had limitations, such as the absence of a Robust dataset, inadequate prediction accuracy, and the lack of cross-validation methods. Hasan et al. (2020) predicted student performance in higher education with the help of video learning analytics and data mining techniques. They contributed to understanding how different classification algorithms can predict students based on their interactions. They used Random Forest, Logistic Regression, Naive Bayes, and K-nearest neighbors (KNN). The random forest algorithm is the best model in their study with 88.3% accuracy. Their limitation is they did not tune the algorithms to improve the accuracy. Al-Zawqari et al. (2022) developed a flexible framework for predicting students' academic performance in online courses without relying heavily on feature engineering. They proposed a new framework that eliminates the feature engineering step, uses model interpretation methods for feature selection, and tests the framework with artificial neural networks and random forests. They utilize the Open University Learning Analytics dataset (OULAD). Their approach is notable for not using traditional feature engineering. Instead, it relies on raw data and model interpretability for feature selection. They used artificial neural networks and random forests. The models achieved high prediction accuracy (86% and 88%) for at-risk students and 93% for predicting drop-out students. Jiao et al. (2022) developed an AI-enabled model to predict student academic performance in online engineering education. They used data from an online engineering course. They used Genetic Programming (GP), a branch of evolutionary computation, to develop the prediction model. The GP model showed an effective prediction of student performance. The GP model was identified as the most effective approach due to its ability to model complex relationships in the learning data. Syed Mustapha (2023) predicted student performance using various data mining techniques and feature selection methods. They used the OULA dataset. They use machine learning models like the Ensemble Model, XGBoost, Neural Networks, Decision Tree for classification, Linear Regression, Support Vector Regressor, Random Forest Regressor, and Gradient Boosted Regressor for regression. Their limitation is that their algorithm needs to be tuned to improve accuracy.

Material and method

This section presents the research design and procedures used to examine various aspects of student exam performance, as outlined in the research

objectives. Visual representations of the overall study and the methodology of the current study are shown in Figure 1. The figure represents the overarching framework that informs this research, showing the essential components and their interrelationships, and it also provides a detailed explanation of the experimental process of this study.

Dataset overview

The number of hours of study dedicated by a student is represented by the feature "Study Hours" and is expressed in numerical form. The "Previous Exam Score" feature is a numerical representation of the student's score in their previous exam. The binary feature "Pass/Fail" is the target feature, with a value of 1 indicating a pass and 0 indicating a failure in the current exam.

The dataset consists of data from 500 students, providing a comprehensive and diverse range of study patterns and previous exam performance results. The dataset used in this study was collected from MrSimple07 (2024). There were two features in the dataset, such as study hours and students' previous exam scores. The chi-square test was performed to investigate the relationship between two variables, and the results indicated a highly significant association, as demonstrated by a chi-square statistic of 91.50 with 1 degree of freedom $\chi^2(1) = 91.50$, $p < 0.001$. The extremely low p -value ($p = 1.12\text{e-}21$) strongly suggests that the null hypothesis of independence between the variables should be rejected.

Additionally, the purpose of the dataset is to forecast a student's exam success or failure based on two primary variables: the number of study hours they have completed and their exam score from the prior exam. The "Study Hours" feature is a numerical variable that shows how many hours a student spends getting ready for the test. Additionally numerical, the "Previous Exam Score" represents the student's performance on their most recent exam. The binary goal variable, "Pass/Fail," has a value of 1 denoting a pass and a value of 0 denoting a failure on the current exam. A robust prediction model is supported by the dataset, which includes data on 500 students' study habits and historical exam results.

Moreover, the dataset focuses on predicting student exam performance using two key features: Study Hours (hours spent preparing) and Previous Exam Scores (prior exam results). A chi-square test confirmed a strong association between these features and the pass/fail outcome. However, potential biases arise as only these two factors are considered,

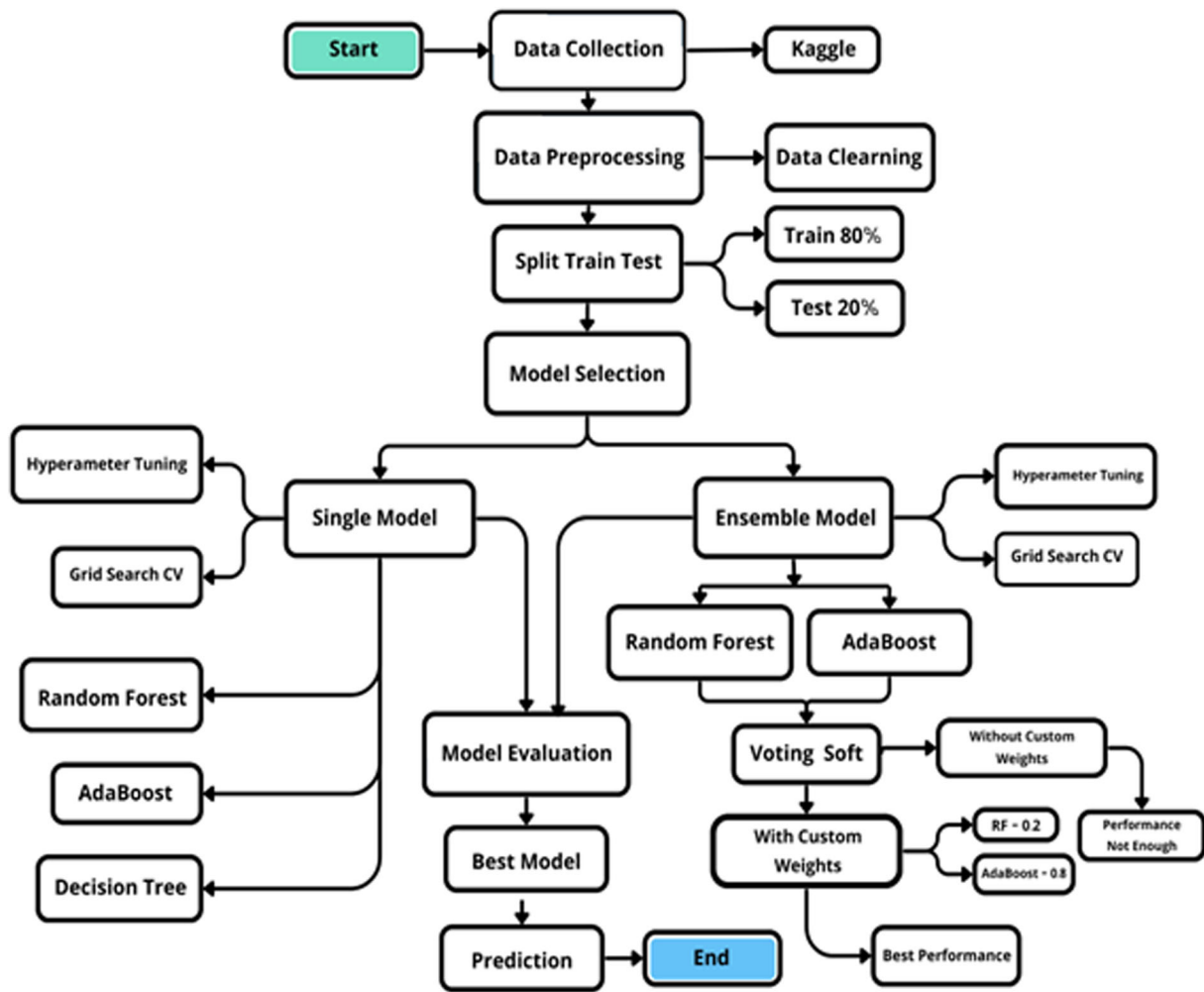


Figure 1. Overview of this study.

excluding other influential variables like motivation or socioeconomic background. The dataset's limited scope could lead to oversimplified predictions, missing the complexity of real-life student performance.

Data clearing

Finding and fixing mistakes in the data set is a part of data cleaning. To remove the rows with NULL values, the `dropna()` method of the pandas data frame library in Python was utilized. However, the dataset does not contain a NULL value. However, the `dropna()` function was employed as a precautionary measure to ensure the absence of missing data. This step verifies the dataset's integrity prior to proceeding with subsequent analysis. Therefore, no features were dropped because there was a high correlation (Figure 2) between each feature, which was important to analyze for this study. However, Pandas data frame `data.isnull().sum()` method is used to check the missing values for each of the individual features in the

dataset, which ensures the dataset has zero null values. An easy way to comply with the conference paper formatting requirements is to use this document as a template and simply type your text into it.

Additionally, The dataset was thoroughly examined for missing values using the `data.isnull().sum()` function from the pandas library. Any rows containing missing data were removed using the `dropna()` function. Following this process, the dataset was confirmed to contain no missing values, thereby ensuring that only complete data were utilized for subsequent analyses.

Moreover, Outliers in the numerical features, namely Study Hours and Previous Exam Scores, were identified through visual inspection using boxplots and distribution plots. Given the characteristics of the dataset, no extreme outliers were detected that required removal or treatment, as their presence did not significantly affect model performance.

Plus, to account for differences in scale between the features, both Study Hours and Previous Exam

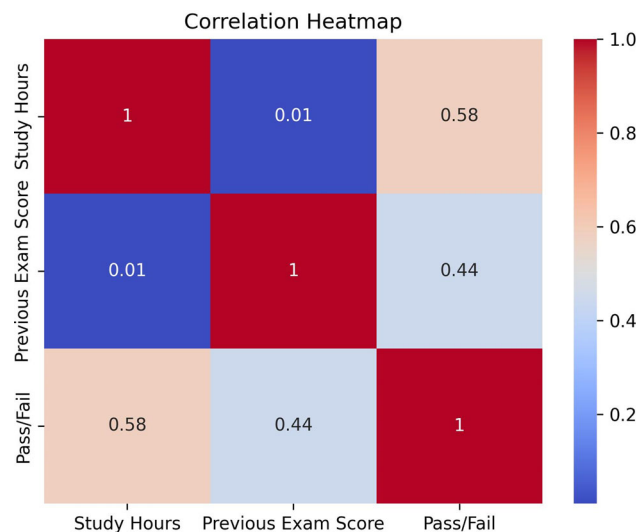


Figure 2. Correlation Heatmap of this study.

Scores were normalized using min-max normalization. This procedure was essential to ensure that the machine learning model treated each feature equitably and that no feature disproportionately influenced the model due to a higher numerical range.

In addition, the dataset was subjected to a thorough verification process to confirm the accuracy of data types. Both input features were confirmed to be numerical, while the target variable, Pass/Fail, was verified as binary. Any discrepancies in data types were rectified to maintain consistency across the dataset.

Besides, a correlation matrix was generated to assess multicollinearity between features. The analysis revealed that the features of Study Hours and Previous Exam Scores were sufficiently independent, allowing both to be used as predictors without introducing redundancy or multicollinearity issues.

Additionally,

Model train test

In this study, 80% of the data were used for training and 20% for testing. However, after splitting the training data, the shape of the dataset was X train shape (400, 2), X test shape (100, 2), y train shape (400, 1), and y test shape (100, 1). The target feature of this dataset is either passing or failing for students. The balanced synthetic dataset (MrSimple07, 2024) is used in this study to predict students' exam performance more precisely. After splitting the train and test data, the model selection step was employed. The purpose of model selection is to identify a model that aligns with our specific needs and criteria, including performance and robustness, rather than trying to locate

the most superior model available. Different models exhibit varying performance depending on factors such as the nature of the data accessible and the kind of predictive problem being addressed. Moreover, these method is used to evaluate and compare various models in order to determine which one is most appropriate for addressing a particular issue. It is important to conduct thorough assessments and comparisons when selecting a model, as this can help ensure that the chosen model is well-suited to the task at hand.

Model development

Individual model

Decision tree classifier. Decision tree classifiers have proven to be effective in a wide range of fields. Their primary advantage lies in their ability to extract descriptive decision-making knowledge from the provided data. These classifiers can be constructed using training sets (Fahad et al., 2023; Hossain et al., 2024). Moreover, a standalone decision tree exhibits a propensity for overfitting, particularly when the tree structure becomes excessively deep. In contrast, the Custom Ensemble model utilizes multiple models to mitigate overfitting, thereby enhancing generalization. While decision trees offer interpretability, they lack the robustness characteristic of ensemble methods. Although decision trees are easy to interpret, they have a tendency to overfit, especially with deep trees. To mitigate this, the Custom Ensemble model incorporates other algorithms to reduce overfitting. The hyperparameters used for Decision Tree include max_depth, which controls the depth of the tree, and

`min_samples_split`, which sets the minimum number of samples required to split an internal node.

Random forest classifier. Random forest is a widely-utilized machine learning algorithm that is trademarked by Leo Breiman and Adele Cutler, which integrates the outcomes of numerous decision trees to arrive at a single conclusion. Its user-friendly nature and adaptability have contributed to its widespread adoption, as it effectively tackles both classification and regression challenges (Fahad et al., 2023a, 2023b). Moreover, the custom weighting (0.8 for AdaBoost and 0.2 for Random Forest) was derived from performance during validation, emphasizing AdaBoost's superior handling of misclassified instances while maintaining the robustness of Random Forest. The soft voting technique integrates these predictions by calculating a weighted average of the predicted probabilities, thereby enabling the ensemble to leverage the strengths of both models. This combination results in improved generalization, enhanced evaluation metrics, and a more favorable trade-off between bias and variance in the final model. Therefore, Random Forest is an ensemble method composed of multiple decision trees, where each tree is built from a subset of the data and features. The model reduces overfitting by averaging the predictions of these trees, which makes it more robust compared to individual decision trees. However, it may not perform well on misclassified instances as AdaBoost does. The hyperparameters used for Random Forest include `n_estimators` (the number of trees in the forest) and `max_features` (the number of features to consider when splitting a node). In this study, Random Forest's strength lies in its ability to generalize well to unseen data, especially in combination with AdaBoost.

AdaBoost classifier. AdaBoost (AB) is a machine learning method known as adaptive boosting. It functions by integrating various algorithms to improve the performance of the classifier. AB demonstrates remarkable sensitivity to noise and outliers in the data. In certain situations, it exhibits reduced susceptibility to overfitting compared to other learning algorithms. Widely regarded as the superior out-of-the-box predictor (Ali et al., 2023). Additionally, AdaBoost emphasizes the improvement of weak learners (typically decision trees) through sequential weight adjustments based on misclassified samples. It demonstrates efficacy in handling difficult-to-classify instances but exhibits sensitivity to outliers. The Custom Ensemble model addresses this limitation by

integrating the strengths of both AdaBoost and Random Forest, mitigating overfitting while concurrently improving performance on challenging cases. Moreover, Random Forest, being an ensemble of decision trees, excels in reducing overfitting through bagging (sampling both data and features). However, it lacks the capacity to focus on difficult-to-classify instances in the manner of AdaBoost. The Custom Ensemble model bridges this gap by combining Random Forest's robustness with AdaBoost's targeted focus on misclassified data. AdaBoost, short for Adaptive Boosting, is an ensemble method that improves the performance of weak learners (in this case, decision trees) by adjusting the weights of misclassified instances at each iteration. The model gives more attention to difficult-to-classify instances by increasing their weight. Although AdaBoost handles such cases effectively, it is sensitive to outliers. In this study, the AdaBoost classifier was optimized using `n_estimators` (the number of weak learners) and `learning_rate` (the contribution of each classifier). The model plays a crucial role in enhancing the overall predictive accuracy, particularly in handling difficult cases.

Ensemble model

An ensemble is a combination of two or more models in which the individual performance score is taken to evaluate the result. In this study, the proposed ensemble model is the combination of two base models, such as a random forest and an AdaBoost model in Figure 3. A soft voting technique is used for the proposed model. Random Forest is effective at reducing variance and overfitting by aggregating multiple decision trees, each trained on a random subset of the data. On the other hand, AdaBoost focuses on improving the model accuracy by iteratively adjusting the weights of misclassified samples, thus emphasizing difficult-to-classify instances. When combined, Random Forest and AdaBoost can mitigate each other's weaknesses while capitalizing on their respective strengths, resulting in a more robust and accurate ensemble model. Additionally, the diversity brought by integrating models trained using different algorithms and techniques enhances the ensemble's ability to capture complex patterns in the data, leading to improved overall performance. Moreover, the standard ensemble employing soft voting averages the probability outputs of both Random Forest and AdaBoost. However, it does not dynamically adjust the influence of each model. In contrast, the proposed Custom Ensemble model employs a custom soft voting mechanism that

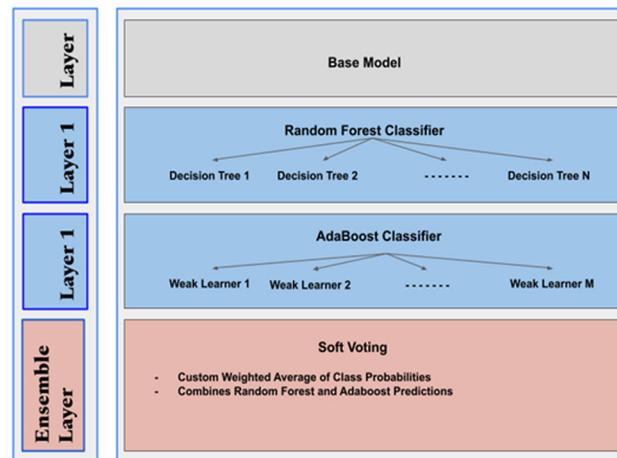


Figure 3. Ensemble model layers of this study.

adjusts the voting weights based on model performance on the validation set. This approach enables the model to assign greater weight to the model that demonstrates superior performance in specific cases, resulting in improved evaluation metrics. Additionally, soft voting combines the probability estimates of multiple classifiers (such as Random Forest and AdaBoost) and averages them to make a final prediction. While standard soft voting equally weights the classifiers, the Custom Ensemble model introduced in this study assigns weights based on model performance on a validation set. The parameters used in the custom ensemble are the voting weights, where AdaBoost is given a higher weight (0.8) due to its superior performance in handling misclassified data, while Random Forest is assigned a lower weight (0.2) for its generalization capabilities. However, the architecture of the proposed model is shown in Figure 3.

Proposed model

In this study, we employed a custom-weighted soft voting ensemble machine learning model comprising AdaBoost and Random Forest to predict the students' exam performance. Initially, we import the necessary Python libraries for data preprocessing, model construction, and performance evaluation. We then define our base models and establish a hyperparameter grid for each. Using a 10-fold cross-validation, we performed a grid search technique to tune the hyperparameters of both AdaBoost and Random Forest classifiers. We then train our tuned models. We incorporate our optimally tuned AdaBoost and Random Forest models to construct an ensemble model using a custom-weighted soft Voting Classifier and assign custom weights to our classifiers (0.8 for AdaBoost and 0.2 for Random Forest). Finally, this

ensemble model is then trained and used for prediction (Figure 4).

Evaluation metrics

Performance evaluation of a classification model involves the use of various metrics, including accuracy, precision, recall, and F1-score. While precision analyzes the accuracy of the model in recognizing positive cases, accuracy evaluates the overall soundness of the model's predictions. A balanced score between precision and recall is provided by the f1-score (Ali et al., 2023; Saheed, 2023). Recall assesses the model's capacity to catch all positive events. Combining these criteria allows for a more thorough knowledge of the model's categorization abilities and a more sophisticated evaluation of its performance. Mathematically (Powers, 2020), defined as

$$\text{Accuracy} = \frac{\text{Number of correct Predictions}}{\text{Total number of predictions}} \quad (1)$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (2)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (3)$$

$$\text{F1 Score} = \frac{2(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (4)$$

Confusion matrix

The confusion matrix (Figure 5) reveals that the proposed model has achieved an exceptional level of performance. Specifically, it correctly identified 64 positive instances (True Positives) and 35 negative instances (True Negatives). The model made only one error by incorrectly classifying a positive instance as negative (False Negative), and it did not make any False Positive

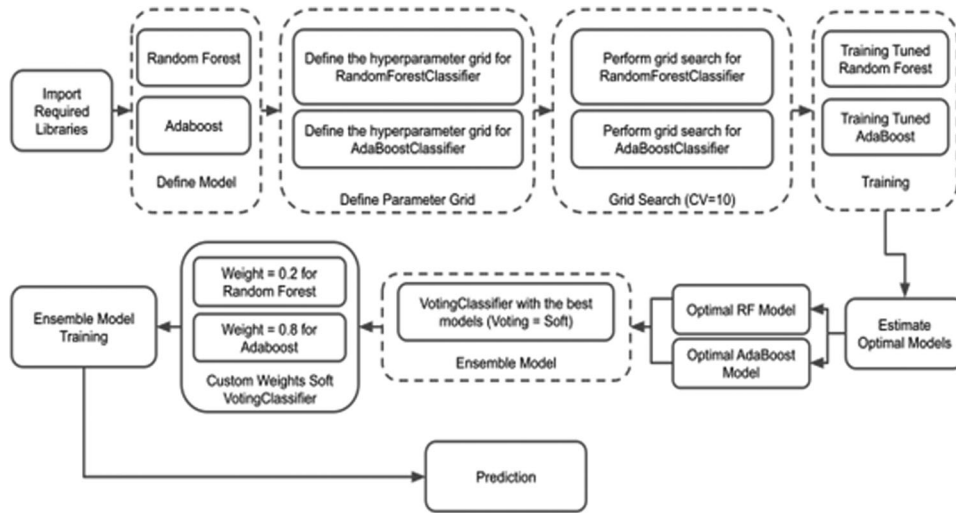


Figure 4. Proposed custom-weighted soft voting ensemble model.

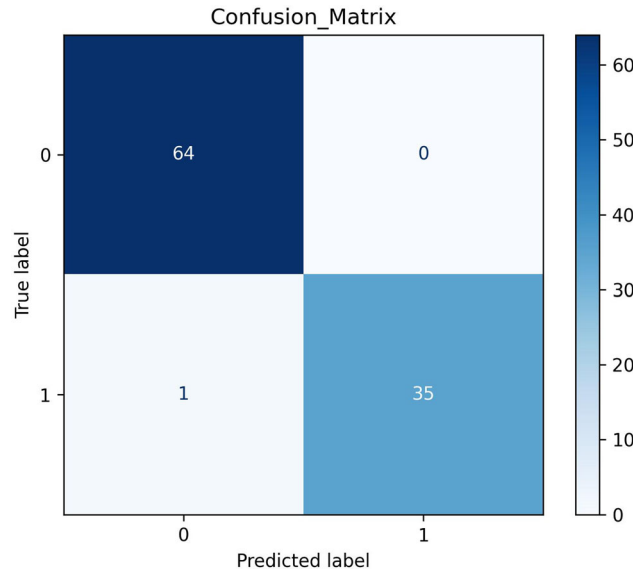


Figure 5. Confusion matrix of this study.

errors. These results indicate a high degree of accuracy, precision, and recall, suggesting that the model is both dependable and stable in its predictions.

Cohen's kappa coefficient

Cohen's kappa statistic measures agreement between two raters and is mostly used to test inter-rater reliability (Metrics and scoring: quantifying the quality of predictions, 2025).

In most correlation statistics, kappa can range from (−1) to (+1). The calculation was considered based on the confusion matrix (Figure 4) in this study. The equation for Cohen's kappa coefficient is as follows:

$$K = \frac{P_o - P_e}{1 - P_e} \quad (4)$$

where P_o is the observed agreement and P_e is the expected agreement.

$$\begin{aligned} P_o &= \frac{\text{Sum of Diagonal Elements}}{\text{Total Number of Instances}} \\ &= (64 + 35)/100 \\ &= 99/100 \\ &\approx 0.99 \end{aligned} \quad (5)$$

$$\begin{aligned} P_e &= \frac{\sum (\text{Row Total} \times \text{Column Total})}{\text{Total Number Of Instances}^2} \\ &= ((64 + 0) \times (64 + 1) + (1 + 35) \times (1 + 0))/(100)^2 \\ &= ((64 \times 65) + (36 \times 1))/(100)^2 \\ &= (4160 + 36)/10000 \\ &= 4196/10000 \\ &= 0.4196 \end{aligned} \quad (6)$$

Evaluating Cohen's Kappa Score:

$$k = (0.99 - 0.4196) / (1 - 0.4196) \\ \approx 0.979 \\ \approx 0.98$$

A high degree of agreement between the actual outcomes and the model's predictions is indicated by Cohen's Kappa score of 0.98, which indicates outstanding performance. Furthermore, the outcome shows almost perfect agreement for this particular study.

Experimental setup. This study made use of the following instruments:

- Google Colab: The only programming language available in a web browser was Python, which Google provided to the scientific community (Kappa Statistics – An Overview | ScienceDirect Topics, 2024). The application was used in conjunction with the following Python libraries:

A flexible Python package for working with multi-dimensional arrays and matrices is called NumPy (Google Colaboratory, 2024).

- Matplotlib: This cross-platform package was used to speed up data visualization and graphing when used with NumPy (2024).

Moreover, Data was processed and analyzed in tabular form using a software library called Pandas (Matplotlib—Visualization with Python, 2024).

Among the many tools available for using machine learning algorithms are those for regression and classification, which are part of the extensive

software library Scikit-learn (Ahmed et al., 2024; Pandas – Python Data Analysis Library, 2024; Scikit-Learn: Machine Learning in Python, 2024).

Result and discussion

The study comprised the synthetic data of 500 students to predict their upcoming exam performance. However, the study found that students who obtained scores ranging from 40 to 100 and 40 to 60 in their previous exam and dedicated 0–5 or 5–10 hours of study, respectively, for the upcoming exam, had a higher likelihood of failing. In contrast, students who scored between 60 and 100 in their previous exam and devoted 5–10 hours of study to the upcoming exam had a higher chance of passing (Figure 6).

Moreover, Table 1 in this study presents the evaluation metrics clearly, demonstrating that the proposed customized ensemble model outperformed the single models such as (Decision Tree, Random Forest, and AdaBoost) in terms of the area under the ROC curve, and Table 2 presents the training and prediction time for the baseline and proposed customized ensemble model.

Table 1. Evaluation metrics of this study.

Model	Accuracy	Precision	Recall	F1-score	ROC
Decision tree	0.990	0.99	0.99	0.99	0.99
Random forest	0.990	1.0	0.972	0.986	0.998
AdaBoost	0.964	1.0	0.898	0.946	0.977
Ensemble model	0.990	1.0	0.972	0.986	0.998
Custom ensemble model	0.991	0.99	0.99	0.99	1.0

Table 2. Training and prediction time of this study.

Model	Training time(Sec)	Prediction time(Sec)
Ensemble model	4.688	0.279
Custom ensemble model	0.163	0.015

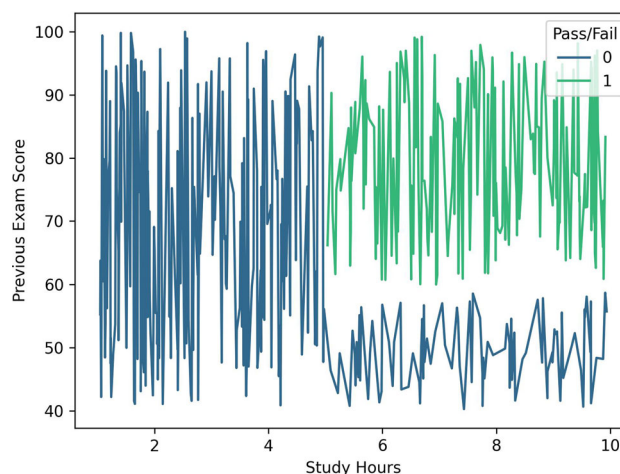


Figure 6. Study hours vs previous exam score by pass/fail.

It is evident from the table that the proposed model's prediction time is faster than that of the baseline ensemble model. On the other hand, the results of Figure 7 indicate a high level of consistency and robustness in the performance classification model, as evidenced by a mean accuracy of 99.75% across 10-fold cross-validation. This validation procedure entails dividing the data set into 10 subsets, using 9 for training and 1 for testing, and repeating the process 10 times. The mean accuracy of 99.75% obtained from this process indicates that the model correctly predicts the class labels for 99.75% of the instances in the dataset, on average. This high accuracy suggests that the model is well generalized and can make accurate predictions on unseen data. Researchers can view this as a strong indication of the model's overall effectiveness in capturing patterns and relationships within the data.

The data reveals that, on average, students allocate roughly 5.49 hours to studying, with a median study time of 5.62 hours. However, the range in study hours is considerable, ranging from 0 to 8.89 hours. The interquartile range (IQR) for previous exam scores is 52.73. The skewness values indicate a

slight leftward skewness for study hours and exam scores while pass/fail data exhibit positive skewness, suggesting a tendency towards more passing grades. Additionally, the negative kurtosis values indicate a relatively flat distribution for all three variables. Overall, the study hours and exam scores exhibit moderate variability with some asymmetry, while pass/fail data display a right-skewed distribution (Figure 8).

The graph (Figure 9) depicts the decision boundaries for Class A, represented by blue points, and Class B, represented by red points. The decision boundaries, depicted by solid lines, illustrate a well-defined separation between the two classes without any overlap. This separation indicates that the model effectively distinguishes between the classes and follows a clear pattern. Furthermore, the absence of overlap between the two classes suggests a high level of accuracy in the model's predictions, with distinct regions assigned to each class. The decision boundaries demonstrate the model's ability to capture the underlying patterns in the data, as evidenced by the significant role played by features such as previous exam scores and current study

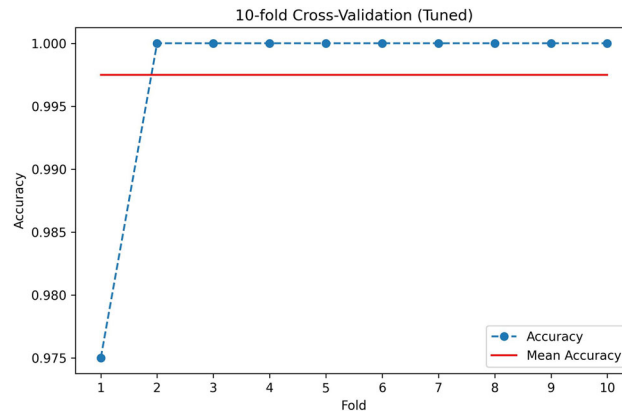


Figure 7. 10-fold cross-validation of this study.

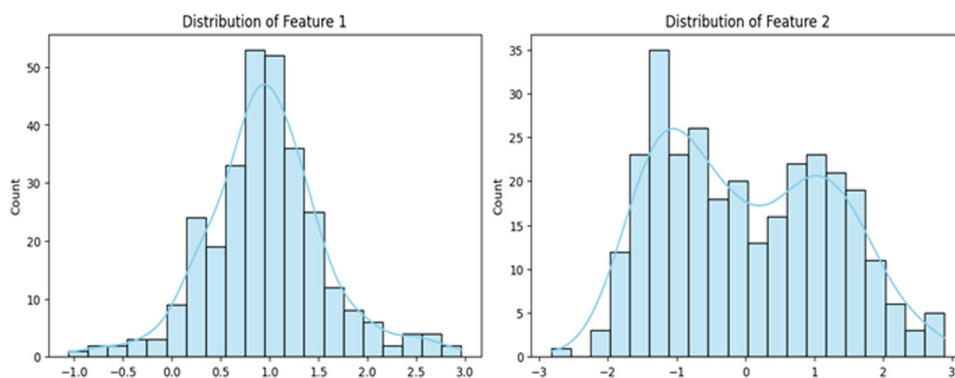


Figure 8. Data distribution.

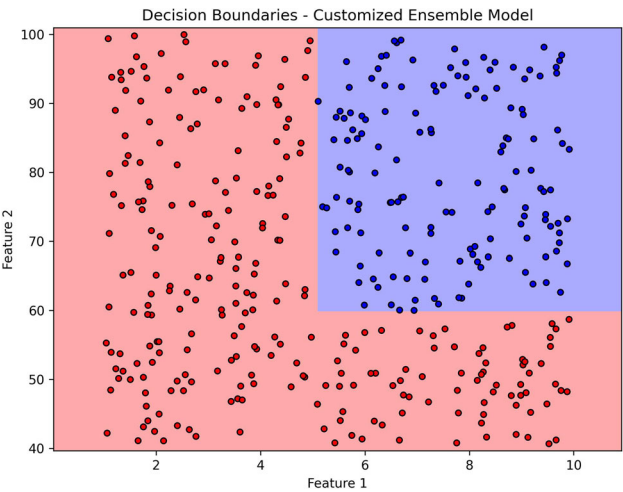


Figure 9. Decision boundaries of customized ensemble model.

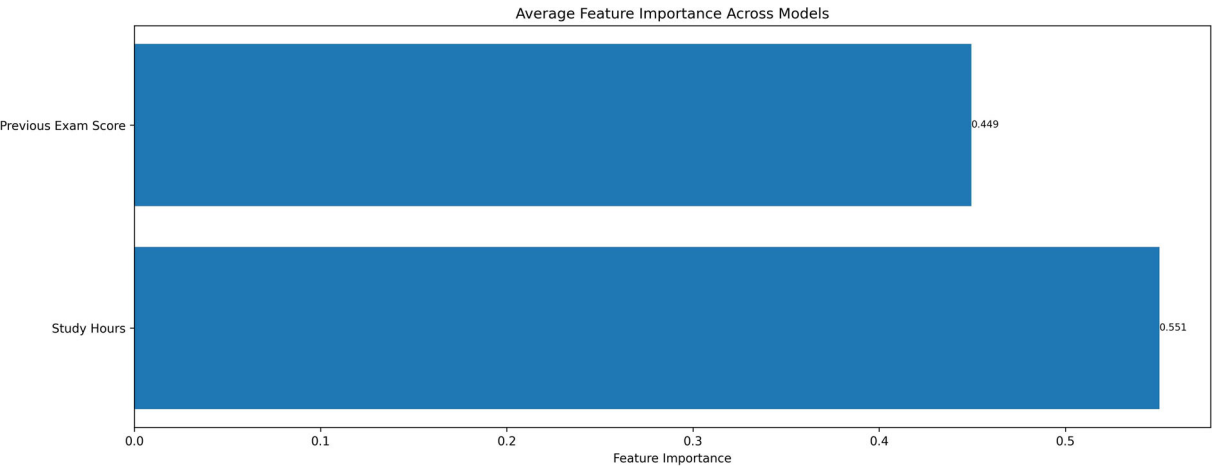


Figure 10. Important features identified in the dataset.

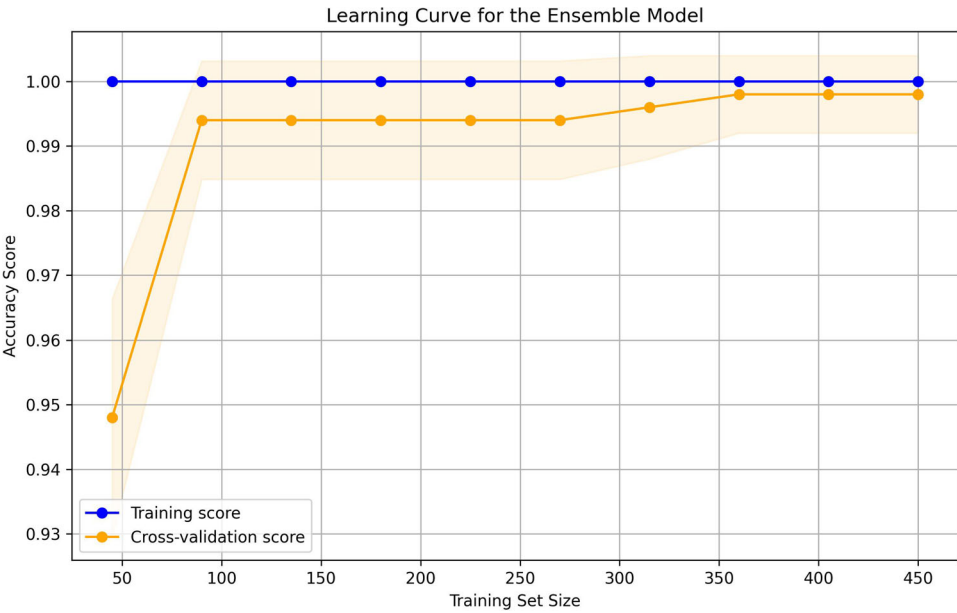


Figure 11. The learning curve of this proposed ensemble model.

hours. Overall, the decision boundaries reflect a robust classification performance, clearly demarcating Class A from Class B.

The feature importance graph (Figure 10) illustrates that the scores of previous exams and study hours were 0.45 and 0.547, respectively, indicating that the model places higher importance on the second feature in making predictions. The score of 0.547 suggests that the second feature or current study hours have a relatively stronger impact on the model's decision-making compared to the first feature or previous exam score, with a score of 0.45.

The learning curve (Figure 11) indicated that the model demonstrated strong performance across the ten iterations, with cross-validation scores ranging from 0.94 to an impressive 0.9975. It is noteworthy that the training score remained steady throughout the iterations, which signifies that the model quickly grasped the patterns present in the training data

and memorized them effectively. Although the lack of improvement in the training score may suggest that the model has reached its capacity for learning from the training set, the consistently high cross-validation scores suggest that the model can generalize well to unseen data. This stable performance on the validation set is a positive indication that the model has acquired knowledge effectively and can accurately predict outcomes on new, unseen data.

Nonetheless, the ROC curve value for this investigation was 1.00, indicating (Figure 12) the excellent performance of the binary classification model and faultless discrimination between classes. This result translates to a 100% true positive rate (sensitivity) and a 0% false positive rate (specificity), indicating that the model can reliably distinguish between positive and negative cases while avoiding incorrect predictions. This finding highlights the model's best discriminatory power and demonstrates how consistently it can

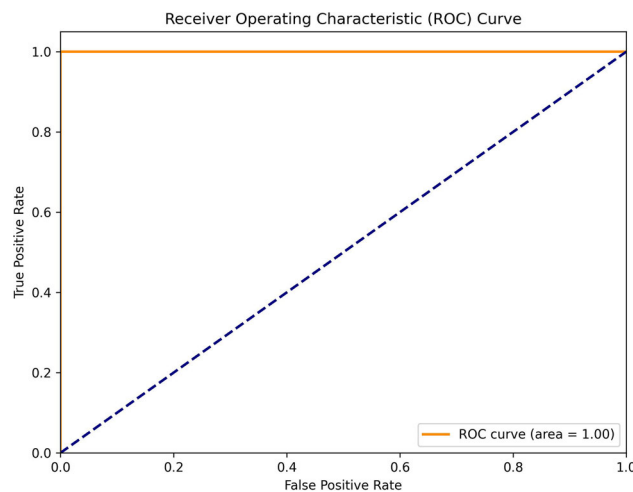


Figure 12. ROC curve of the ensemble model.

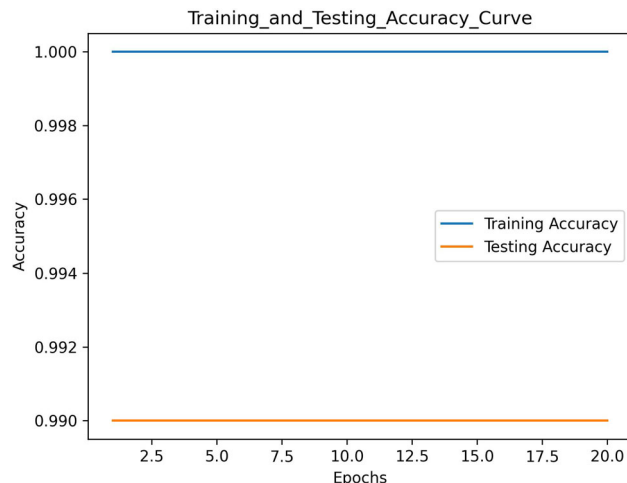


Figure 13. Train-test accuracy of the ensemble model.

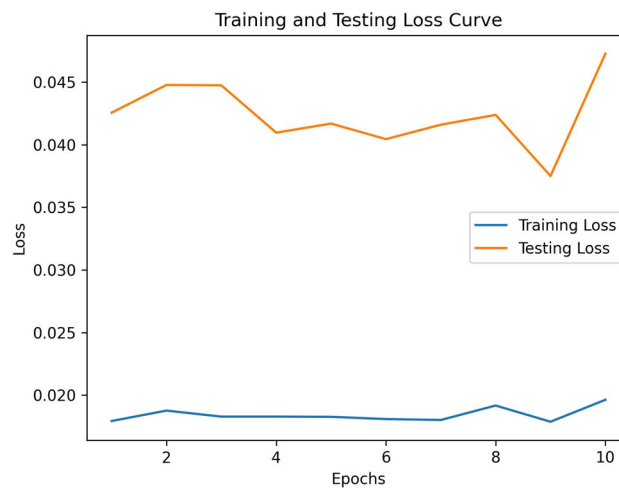


Figure 14. Training and testing the loss of the ensemble model.

Table 3. Comparison between our efficient model with other studies.

Model	Accuracy	Citation
Random forest	88.3%	(Hasan et al., 2020)
Random forest	88%	(Al-Zawqari et al., 2022)
ANN	98.3%	(Alsariera et al., 2022)
Custom ensemble model	99.1%	99%

identify occurrences, underscoring its possible importance in the context of the study.

As can be observed in Figure 13, training accuracy stays constant at 100% while testing accuracy levels out at 99%. This suggests that the model learns almost perfectly on training data but performs slightly worse on testing data that hasn't been seen yet. The model appears to have successfully retained the training set based on its consistent 100% Training Accuracy. However, the 99% testing accuracy, while still very good, indicates that the model performs well in general but may have some issues with a small number of cases in the testing set. Moreover, Figure 14 shows that the testing loss stays below 0.020 accuracy loss, however, the training loss varies between 0.035 and 0.045 after utilizing 10 epochs.

Comparison between our customized model with other studies

Table 3 represents the comparison between our customized models with other existing studies. Table III shows that our customized model got higher results than the existing studies.

Conclusions

This study proposed a customized ensemble machine learning model that combined Random Forest and

AdaBoost classifiers, achieving superior accuracy in predicting student exam performance. The model not only outperformed traditional approaches but also demonstrated high reliability in early prediction, which could enable timely interventions for at-risk students. However, the study has certain limitations, including the small dataset size and the exclusion of important factors such as socio-economic and psychological variables. Furthermore, the binary classification (pass/fail) restricts the model's applicability to more complex grading systems. To address these limitations, future research should aim to expand the dataset, incorporate more diverse features, and explore deep learning techniques for enhanced accuracy. Additionally, extending the model to multi-class classification and benchmarking it against recent models from 2023 to 2024 will be crucial for validating its robustness and improving its applicability.

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Disclosure statement

There is no conflict of interest between the authors.

Author contributions

RA: model designing, programming, experimental, and manuscript writing, figure drawing. NF: conceptualization, manuscript writing, and manuscript review. MMI: supervision, manuscript writing, and manuscript review. KG: manuscript writing and manuscript review, supervision. MMA: manuscript review and writing. MR: manuscript review and writing. Moreover, all authors have read and approved the final version of the manuscript submitted for publication.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, [Kah Ong Michael Goh, email-  michael.goh@mmu.edu.my] upon reasonable request.

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